STRUCTURED TECHNICAL REPORT

PC Build

Component Selection, Physical Assembly, Windows Setup, and Validation

Hardware | Assembly | Firmware | Drivers | Windows

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Bottom Line Up Front (BLUF)

I built a Windows 11 Pro desktop around an AMD Ryzen 7 9700X, a Gigabyte GeForce RTX 5070 GAMING OC, 64 GB of DDR5 memory, and a 2 TB NVMe SSD. The system reached UEFI, booted Windows, drove my QHD monitor at approximately 180 Hz, and completed a roughly 10-minute combined CPU and GPU load check. Before daily use, I confirmed a firmware and Windows security baseline: fTPM 2.0, Secure Boot with UEFI-only boot, Memory Integrity, and reputation-based protection.

Two problems were caught and corrected along the way. During assembly, I found and replaced the wrong CPU cooler mounting screws and moved the cooler fan cable to the correct header. On first boot, the graphics card produced no picture, so I used the motherboard's HDMI output to reach UEFI and install Windows. After installing the NVIDIA driver and restarting, I confirmed video output from the graphics card.

The report closes with the lessons I took from the build and a component-by-component plan for upgrading the system over time instead of replacing it.

1. Objectives and Component Selection

1.1 Objectives

I wanted a desktop PC that could support full-time work in IT and cybersecurity, a personal home lab, simulations, heavy multitasking, future virtualization, and occasional gaming. I already had a QHD 180 Hz monitor available to use.

I set a budget of roughly $2,500. That was a fair target given my requirements of 64 GB of RAM, 2 TB of storage, and a white case with visible lighting, especially with GPU and RAM prices running particularly high at the time of the build.

1.2 How the Parts List Changed

- **CPU and GPU.** I kept the 9700X and found it on Amazon for $279.99. I considered the white Yeston RTX 5070 and the RTX 5070 Ti. The 5070 Ti listings I found were around $1,000, so I chose the Gigabyte GAMING OC 5070 and confirmed its part number before purchase.
- **Cooling and Chassis.** I compared the Thermalright Phantom Spirit 120 SE ARGB with the ARCTIC Freezer 36 A-RGB. The prices were similar, so I chose the white ARCTIC cooler and the white V100RW case with its four included fans.
- **Storage.** The WD Black SN7100 was the placeholder after I compared it with the SN850X, Samsung 990 PRO, and other drives. I kept the 2 TB requirement for future lab storage and purchased the Predator GM7 when I found it for $278.99.
- **Memory.** I compared G.Skill and TeamGroup options and purchased the memory last. The black Corsair 64 GB DDR5-6000 CL40 kit cost about $110 less than the TeamGroup kit discussed at the time. I kept the capacity and rated transfer rate and accepted the looser CL40 timing and the darker finish.

1.3 Installed Parts

Component	Installed Model and Role	Recorded Purchase
Processor	AMD Ryzen 7 9700X. AM5 socket, 8 cores, 16 threads, and integrated graphics. [1]	Amazon, $279.99
Motherboard	MSI MAG B850 TOMAHAWK MAX WIFI (MS-7E62). ATX board with DDR5, M.2, PCIe, and wired and wireless networking. [2]	Newegg, $209.99
Graphics Card	Gigabyte GeForce RTX 5070 GAMING OC 12G (GV-N5070GAMING OC-12GD). Desktop card with 12 GB of GDDR7 for QHD graphics and display output. [3]	Newegg, $699.99
Memory	Corsair Vengeance RGB (CMH64GX5M2N6000Z40). Two 32 GB DDR5-6000 CL40 modules with AMD EXPO support. [4]	Newegg, $879.99
CPU Cooler	ARCTIC Freezer 36 A-RGB White (ACFRE00125A). Tower heatsink with two 120 mm fans in a push-pull arrangement. [5]	Newegg, $49.19
Storage	Acer Predator GM7 2 TB (BL.9BWWR.119). M.2 2280 PCIe 4.0 x4 NVMe system drive. [6]	Amazon, $278.99
Power Supply	Corsair RM850x 850 W, ATX 3.1, fully modular. Supplies motherboard, CPU, and GPU power. [7]	Newegg, $149.99
Case	Lian Li Vector V100R White (V100RW). ATX mid-tower with three side fans and one rear fan preinstalled. [8]	Amazon, $78.99

1.4 Product Images

Figures 1 through 8 show the selected models as pictured on manufacturer and retailer product pages. Each caption names the image source, which is linked in Appendix A.



Figure 1. *AMD Ryzen 7 9700X retail package. Image from AMD. [1]*



Figure 2. *MSI MAG B850 TOMAHAWK MAX WIFI. Image from the MSI US Store. [2]*



Figure 3. *Gigabyte RTX 5070 GAMING OC 12G. Image from Gigabyte. [3]*



Figure 4. *Corsair Vengeance RGB, CMH64GX5M2N6000Z40. Image from Newegg. [4]*



Figure 5. *ARCTIC Freezer 36 A-RGB White, ACFRE00125A. Image from Newegg. [5]*



Figure 6. *Acer Predator GM7 2 TB, BL.9BWWR.119. Image from Jumbo Computer. [6]*



Figure 7. *Corsair RM850x 850 W, ATX 3.1. Image from Corsair. [7]*



Figure 8. *Lian Li Vector V100R White, V100RW. Image from Newegg. [8]*

2. Compatibility and Preparation

2.1 Fit and Connections Checked Before Assembly

Check	Result
Platform and Memory	The 9700X and the B850 motherboard both use AM5. The board uses DDR5, and a two-module installation uses slots DIMMA2 and DIMMB2. [1, 2]
Cooler Height	ARCTIC lists the cooler at 159 mm, and Lian Li allows up to 178 mm. The nominal height margin was 19 mm. [5, 8]
Graphics Card Length	Gigabyte lists the card at 327 mm, and Lian Li allows up to 420 mm. The nominal length margin was 93 mm. The 16-pin power plug and its cable also needed room in front of the side panel. [3, 8]
PSU Fit and Capacity	The 160 mm modular PSU fit the case's ATX bay, which accepts units shorter than 220 mm. Its 850 W rating exceeded Gigabyte's 750 W recommendation for the card. [3, 7, 8]
Storage and Expansion	The Gen4 M.2 2280 drive fit M2_1, a CPU-connected PCIe 5.0 x4 slot. The board has four M.2 slots; M2_3 shares lanes with the PCI_E3 expansion slot (Section 11.3). [2, 6, 13]

2.2 Build and Recovery Media

I reviewed build videos and waited for the RAM to arrive so I could assemble and check the system in one session. I prepared two 32 GB USB drives. One was a Windows 11 installer I created with Microsoft's Media Creation Tool, which labels the drive ESD-USB. The other held the MSI manual and five driver ZIP files in labeled folders, including a wired-network driver kept as a recovery package. [10]

3. Physical Assembly

3.1 CPU, Memory, and SSD on the Bench

I began assembly on August 24, 2026, with all components laid out on a table.

1. **CPU.** With the motherboard out of the case, I opened the AM5 load plate, lined up the triangle on the CPU with the triangle on the socket, set the 9700X into the keyed socket, and closed the load plate and retention lever. The alignment marks fix the orientation, so the CPU settles into place without force. [2]
2. **Memory.** I installed the RAM before the SSD. I aligned the keyed notches, pressed the two 32 GB DIMMs into A2 and B2, and checked that the retention tabs closed. These are the board's specified slots for a two-module configuration. [2]
3. **SSD.** I removed the heatsink over the top M.2 slot, M2_1, and inspected the tool-free retention mechanism after finding no separate retaining screw. I inserted the GM7 at an angle, lowered the free end into the retention latch, and reinstalled the MSI heatsink. I left the SSD's factory label in place and peeled the protective film from the heatsink's thermal pad before locking the heatsink down. [2]

3.2 Cooler Installation and Screw Correction

I attached the cooler mounting brackets at the socket, applied thermal compound, and mounted the Freezer 36 with its fans oriented to move air through the tower toward the rear exhaust fan. The detachable fans gave me access to the heatsink screws. The cooler uses separate leads for fan power and lighting: a 4-pin PWM cable for the fans and a 5 V addressable RGB cable for the LEDs. [5]

After the motherboard was in the chassis, I noticed that one side of the cooler mount sat higher than the other. Retightening the brackets did not level it. I removed the fans to inspect the hardware and found that I had used the Intel screws on the AMD mount. I replaced them with the AMD screws, rechecked the brackets, and confirmed the cooler was secure. The cooler shifted slightly during this work but remained seated against the CPU. The later thermal check (Section 9) served as the functional test of the mount.

3.3 Case, Power Supply, and Motherboard

I removed the case panels, checked the standoffs against the ATX mounting holes, and mounted the RM850x in the bottom bay with its fan intake facing the filtered floor opening. Before installing the motherboard, I routed the two CPU (EPS) cables up behind the motherboard tray and staged the 24-pin cable on the right. I kept the GPU cable low until the card was installed.

With the case on its side, I lowered the board in, aligned the rear I/O and standoffs, and fastened it to the tray. I then seated the 24-pin ATX connector and both CPU power connectors. Routing the CPU cables first kept the connectors along the board's top edge accessible after the tower cooler was in place.

3.4 Front Panel, Fans, and Lighting

I connected the front-panel cables before installing the graphics card, while the headers along the board's lower edge were still easy to reach. The front-panel plug went to JFP1, the front USB cable to JUSB3, the USB-C cable to JUSBC1, the internal USB 20Gbps Type-C header, and the HD Audio cable to JAUD1.

The cooler and the four case fans each use two circuits. The PWM circuit powers the fan motors and controls their speed, and the separate 5 V ARGB circuit powers and controls the LEDs. The cooler's daisy-chained fans went to CPU_FAN1 and its lighting to JARGB_V2_3. The case fans went to SYS_FAN1 and JARGB_V2_2 as one group, with the rear fan serving as exhaust. [2, 5, 8]

I first plugged the cooler's PWM cable into PUMP_SYS1. I moved it to CPU_FAN1 and verified every header name again during the pre-power inspection, leaving PUMP_SYS1 empty. Figure 9 in Section 3.6 maps the final connections.

3.5 Graphics Card and Pre-Power Inspection

I removed the rear expansion-slot covers for positions 2 through 4 and seated the Gigabyte card in PCI_E1. Because I did not hear the slot latch click, I checked the latch, the evenness of the seating, and the bracket's fit against the case. I installed the bracket screws and replaced the unused fourth cover. I used the RM850x's supplied 16-pin (12V-2x6) GPU cable, checked that it was fully seated, and routed it so the side panel would not press against the connector. The card's free end rests on a support post, which I checked for level and for clearance from the fans. [3, 7]

Before applying power, I checked the motherboard and CPU power connections, the GPU and its cable, the RAM latches, the M.2 heatsink, the front-panel connections, the fan and lighting headers, and cable clearance. Unused modular cables were left out of the system. I then moved the assembled tower to the monitor for first boot.

3.6 Connection Map

Figure 9 shows every connection in the finished build on one page, grouped by circuit. Each line runs from a component to the header or slot it uses, so the figure also works as a checklist for future maintenance and upgrades.

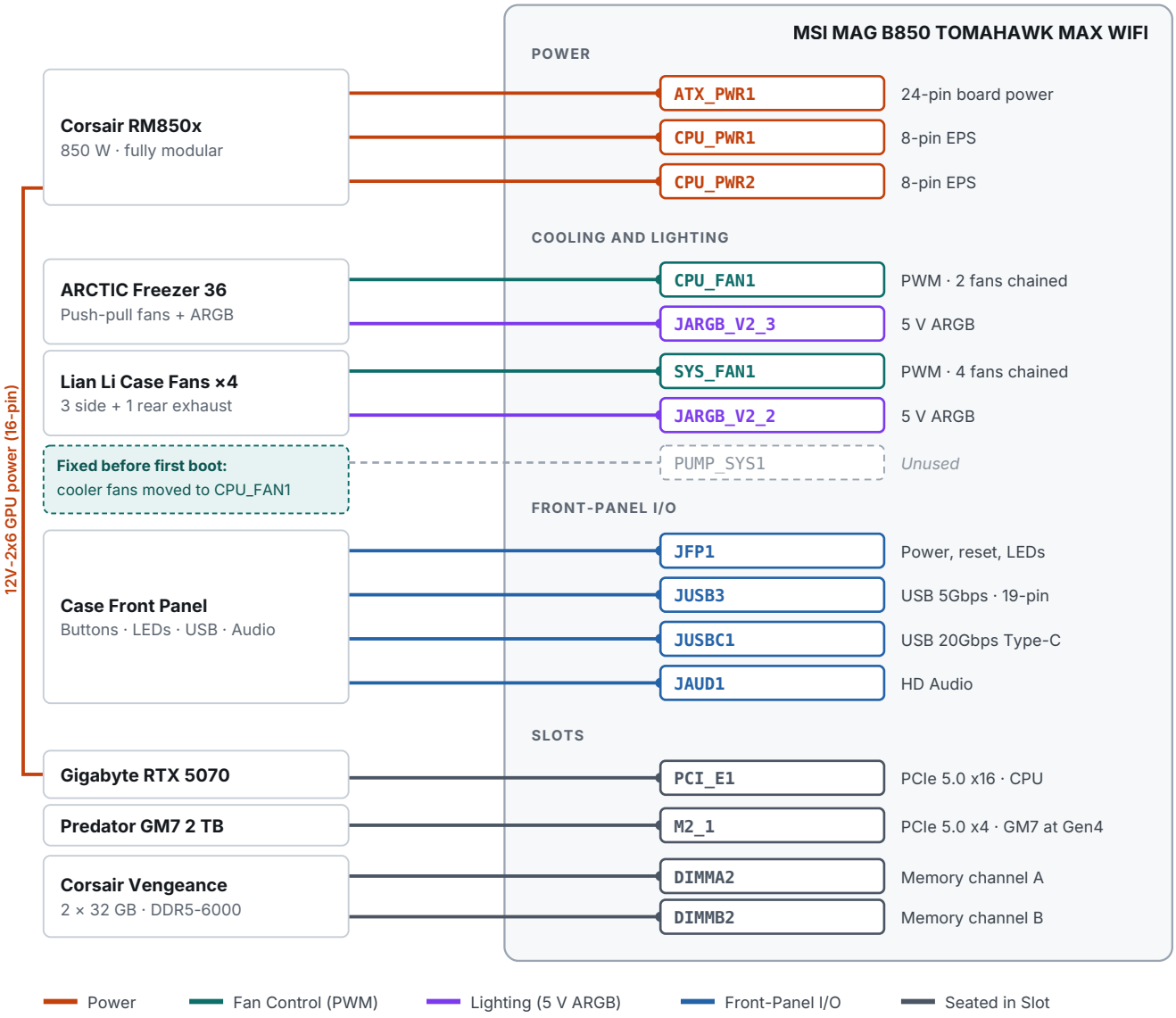


Figure 9. *Connection map of the finished build, color-coded by circuit. Header and slot labels follow the MSI user guide. Schematic, not to scale. [2, 5, 7, 8, 13]*

4. First Boot and UEFI

4.1 No Picture from the Graphics Card

On August 25, the system powered on with fans spinning and lighting active. The green BOOT debug LED was lit, but the monitor stayed blank with DisplayPort connected to the RTX 5070. Switching to the card's HDMI output produced the same result. With no operating system on the SSD yet, a lit BOOT indicator was consistent with the system reaching the boot-device stage. [2]

I moved the HDMI cable to the motherboard and reached MSI UEFI through the Ryzen processor's integrated graphics. This isolated a working display path and let me inspect the platform without disturbing the graphics card. The first UEFI view showed firmware E7E62AMS.2A40 (build date June 19, 2025), 64 GB of RAM at 4800 MT/s, the GM7 drive, and a CPU temperature of about 51 °C. The PCI_E1 link settings were on Auto. The graphics settings showed PEG as the initial display adapter, integrated graphics (UMA) on Auto, and HybridGraphics disabled. I left those values unchanged.

After I installed the NVIDIA driver and restarted, I moved the cable back to the graphics card and the display worked. This restored the intended display path. The evidence does not identify which part of the initial display negotiation caused the blank screen.

4.2 Memory and Boot Settings

Setting	Build Record
Memory Profile	Changed from Standard to EXPO Profile 1; the memory then ran at 6000 MT/s. The first boot at 4800 MT/s provided a default-speed baseline.
PCIe Resources	Re-Size BAR and Above 4G Decoding were already enabled. PEG remained the primary display adapter, and the PCIe link settings stayed on Auto.
Firmware Revision	Kept E7E62AMS.2A40 during setup because it recognized all of the hardware and allowed installation to proceed. A later update is planned in Section 11.2.
CPU Tuning	Precision Boost Overdrive stayed on Auto, MSI Game Boost stayed off, and I made no manual voltage or multiplier changes. The memory profile was the only deliberate performance change at this stage.

Before installing Windows, I also confirmed Secure Boot and UEFI-only boot and enabled SVM under Overclocking > Advanced CPU Configuration > AMD CBS; Section 8.1 explains those settings. I saved the UEFI settings with F10 and proceeded to Windows once memory and storage were recognized. A November 10 hardware inventory later reported BIOS 2.A40 and 6000 MT/s for both memory modules, consistent with the build record.

5. Windows 11 Pro Installation

1. With the monitor connected to the motherboard's HDMI port, I booted the installer USB from the MSI UEFI boot menu, completed the region and keyboard prompts, and entered the Windows 11 Pro product key when Setup requested it. [10]

2. At the destination screen, Setup listed Disk 0 Unallocated Space (approximately 1.9 TB) and Disk 1 ESD-USB (29.8 GB). I selected Disk 0, the blank internal GM7. The installer USB remained available as boot media.
3. At the network screen, I connected Ethernet, set up the computer as a new PC, and signed in with my Microsoft account.
4. At the desktop, I closed the optional MSI utility installer, ran Windows Update first, and then installed the platform and graphics drivers. After the driver checks and a second update pass, I confirmed Windows 11 Pro activation in Settings.

The installation, restart, network, and activation checks established a working Windows baseline. After Windows Update reported the system up to date on August 25, 2026, it was running Windows 11 Pro, version 25H2, OS build 26200.9168, from the August 11 cumulative update (KB5121003). [18] Section 6 lists the driver versions.

6. Driver Installation and Checks

Order and Driver	Package or Source	Package Version	What Happened
1. AMD Chipset	MSI USB, `amd_chipset_drivers_am4_am5.zip`	7.12.04.858	Extracted the package in Windows, ran the installer, and restarted.
2. NVIDIA Display	NVIDIA Game Ready Driver for the desktop GeForce RTX 5070	610.88 WHQL	Selected the desktop RTX 5070 package and used Express installation. Restarted, moved the monitor cable from the motherboard to the GPU, and confirmed the desktop appeared.
3. Qualcomm Bluetooth	MSI USB, `qualcomm_ncm865_bt.zip`	3.1.0.1262	Ran MSISetupDrv64 from the extracted package. Windows then discovered the phone.
4. Qualcomm Wi-Fi	MSI USB, `qualcomm_ncm865_wifi.zip`	3.1.0.1430	Ran the Wi-Fi installer. Device Manager then listed the Qualcomm FastConnect 7800 Wi-Fi 7 adapter.
5. Realtek Audio	MSI USB, `realtek_audio_USB_R.zip` (Realtek HD Universal Driver)	6.4.0.2443	Ran Setup and restarted when prompted.
Ethernet (Backup)	MSI USB, `realtek_pcielan_w11.zip`	10.79.50.1003	Kept as a recovery package. Ethernet already worked during setup, and the Realtek controller appeared in Device Manager.

The MSI package versions are the releases MSI listed for this board's driver files on the day of the build. The NVIDIA driver, Game Ready version 610.88, came directly from NVIDIA. The Ethernet package was kept as a backup rather than installed. [2, 11]

Using the front USB ports for the keyboard and mouse receivers confirmed that the front-panel USB connection worked. Bluetooth discovery was tested separately because those devices used USB receivers. In the screens reviewed, Device Manager showed the NVIDIA and AMD graphics adapters and the expected network adapters without warning icons. Windows Advanced display settings then confirmed the NVIDIA adapter as the active display connection. [2, 11]

7. Fan and Display Settings

7.1 Fan Behavior

After Windows activation, I returned to MSI Hardware Monitor in UEFI. Both fan groups used PWM control with Smart Fan mode. The case fan group initially ran at about 1,700 RPM. I set the SYS_FAN1 curve to 35% at 0 °C, 40% at 40 °C, 50% at 55 °C, 70% at 65 °C, and 100% at 80 °C. The fan screen then showed about 925 RPM, a single reading taken in UEFI.

I set a step-up time of 0.3 seconds and a step-down time of 0.7 seconds, then saved with F10. The goal was to smooth the fans' response to brief CPU temperature spikes. The cooler fans remained on their own header and curve.

7.2 Display Settings

Windows reported the LG UltraGear monitor at 2560 × 1440 and 179.96 Hz, connected to the RTX 5070, with a variable refresh range of 49–180 Hz. NVIDIA Control Panel showed 180 Hz, full-range RGB output, and 8 bits per color channel. I reviewed the Adaptive-Sync controls and confirmed the display configuration was complete. The NVIDIA screen also displayed the monitor's G-SYNC Compatible notice. [9]

8. Security Configuration

Because this PC supports cybersecurity work and a personal home lab, I treated its security settings as part of the build. The baseline starts in the firmware and continues through Windows and its software sources.

8.1 Firmware Security

Setting	State	Why It Matters
fTPM 2.0	Enabled	The processor's firmware TPM provides the hardware root of trust that Windows 11 requires and that BitLocker and Windows Hello use to protect keys. [16]
Secure Boot	Enabled	Allows only boot software signed by trusted keys to run, which blocks unsigned boot loaders. [12]
Boot Mode	UEFI only; CSM disabled	Removes the legacy boot path, which Secure Boot requires.
SVM (AMD-V)	Enabled	Provides hardware virtualization for lab VMs, Hyper-V, and Windows Sandbox. Memory Integrity also relies on it.

8.2 Windows Security

Control	Configuration
Edition	Windows 11 Pro, chosen for BitLocker management, Hyper-V, Windows Sandbox, Local Group Policy, and Remote Desktop hosting.
Memory Integrity	On, under Windows Security > Device security > Core isolation. [14]
Reputation-Based Protection	Recommended protections on, under Windows Security > App & browser control. [15]
Antivirus	Microsoft Defender Antivirus provided real-time protection at build time. I installed Malwarebytes Free from the Microsoft Store as an on-demand, second-opinion scanner.
Drivers and Utilities	Drivers came only from MSI's support page and NVIDIA. I skipped driver-updater utilities and optional motherboard suites such as MSI Center to keep the attack surface and software footprint small.
Software Sources	Applications came from official vendor websites or the Microsoft Store.
Browser	Brave is the primary browser, linked to my other devices through Brave Sync's device-to-device sync chain. Edge remains as the Windows fallback, and Chrome is not installed.
BitLocker	Not enabled. This is a home desktop without a meaningful physical-theft risk, so the Windows Hello PIN is sufficient. BitLocker remains available in Windows 11 Pro if that changes. [17]

8.3 Sign-In Trade-Off

I sign in with a Windows Hello PIN. I chose not to require sign-in when the PC wakes from sleep: I set *If you've been away, when should Windows require you to sign in again?* to Never, then cleared the screen saver's *On resume, display logon screen* checkbox, which had kept the prompt in place. A shutdown, a restart, signing out, or locking the PC with Win+L still requires the PIN.

This is a deliberate convenience trade-off for a desktop in my home. Anyone with physical access while the PC is asleep would return to an unlocked session; locking it with Win+L before stepping away closes that gap.

8.4 Later Change

This section records the configuration at the time of the build. By September 2026, Malwarebytes had become the registered real-time protection and Microsoft Defender Antivirus was running in passive mode. The *Home Lab Configuration* report documents that change in Section 7.1.

9. Thermal and Stability Validation

9.1 Idle and Load Results

On August 26, I recorded an idle baseline and then ran a combined CPU and GPU load for roughly 10 minutes while watching temperatures, power draw, fan response, and throttling. I ran the combined load and monitored the sensors in HWiNFO64. Complete run logs were not saved.

Measurement	Idle	Combined Load
CPU Temperature	About 46 °C current; 48 °C average	About 64–65 °C
CPU Package Power	About 23 W	About 88 W
GPU Core / Hot Spot	About 40 °C / 45 °C	About 56 °C / 64 °C
GPU Power and Fans	Fans at 0 RPM	About 124–136 W; fans at about 1,250 RPM
Thermal Throttling	None (CPU throttling flag: No)	None reported in the reviewed readings

The run finished without a crash, unexpected reboot, or display artifact. The readings support the cooler mount and fan behavior under the applied load. Because GPU power stayed well below the card's maximum during the run, I treat this as an initial operating baseline rather than a full stress test.

9.2 Everyday Sustained Load

While OneDrive downloaded and synced my files, I reviewed HWiNFO64 and Task Manager to see how the system handled a sustained, real-world workload across the SSD, network, memory, and background services.

Measurement	Reading During File Sync
CPU Temperature	About 47 °C current, 52 °C average, and 58 °C maximum; no thermal throttling
CPU Package Power	About 24 W current; 48 W maximum
GPU Core / Hot Spot	About 41.5 °C / 46.5 °C; fans at 0 RPM
Memory	About 12 GB in use (19%); DIMMs at about 40 °C
Predator GM7 SSD	43 °C current; 44 °C maximum
Ethernet	Peak download of about 66 MB/s (about 540 Mbps)
Hardware Errors	WHEA: 0. PCIe: link recovery count of 635, with the receiver, replay, bad TLP, bad DLLP, LCRC, correctable, non-fatal, and fatal error counters all at 0

The recovery count records how often the GPU's PCIe link retrained, which can happen as the link changes power states. Because every integrity counter and WHEA stayed at zero, I treat it as a value to monitor rather than a fault. A rapid climb alongside nonzero error counters, WHEA errors, or display problems would justify reseating the card and checking the 12V-2x6 connection.

10. Finished PC

My photographs show the completed assembly through the glass side panel. The white cooler, the two Vengeance DIMMs, the horizontally mounted Gigabyte card, the support post, and the preinstalled case fans are visible. The five views show the system with lighting off and in four lighting colors.

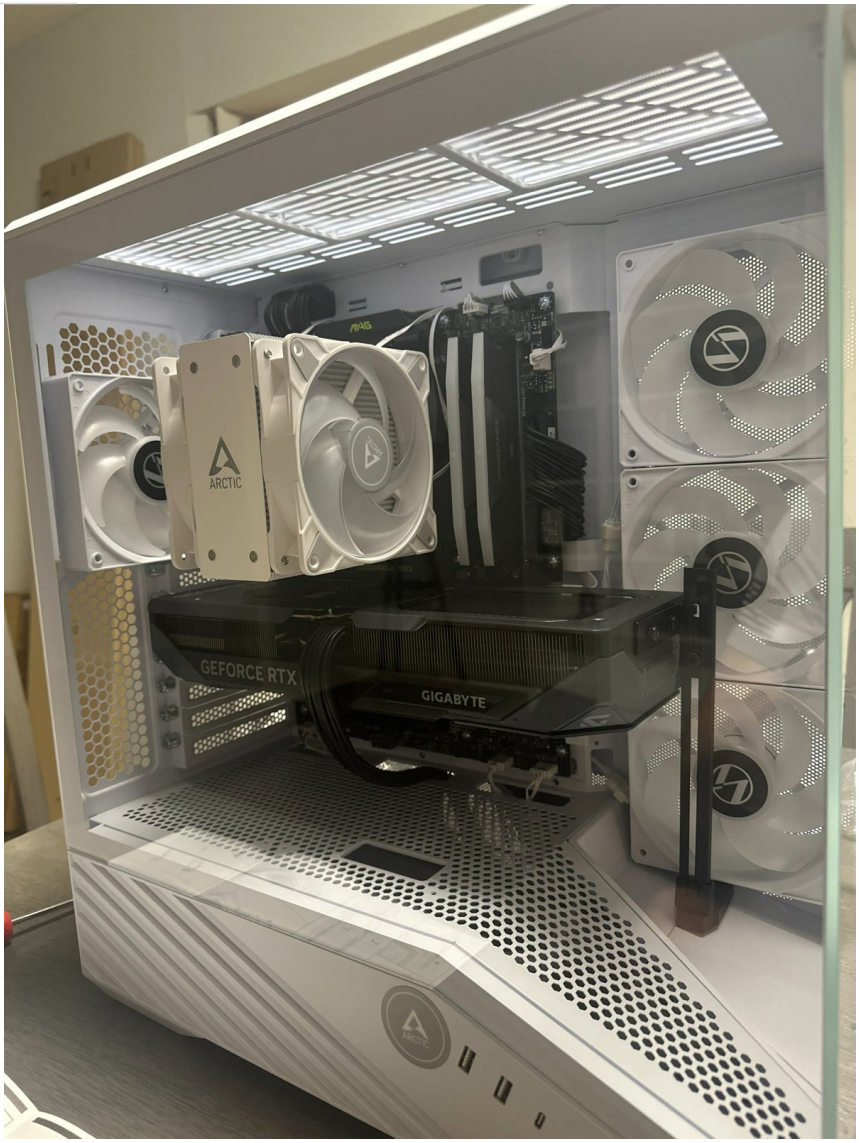


Figure 10. *Finished assembly with lighting off, showing the CPU cooler, both DIMMs, graphics card, support post, and case fans. Photograph IMG_4175.*



Figure 11. *Yellow lighting. Photograph IMG_4176.*



Figure 12. *Purple lighting. Photograph IMG_4177.*

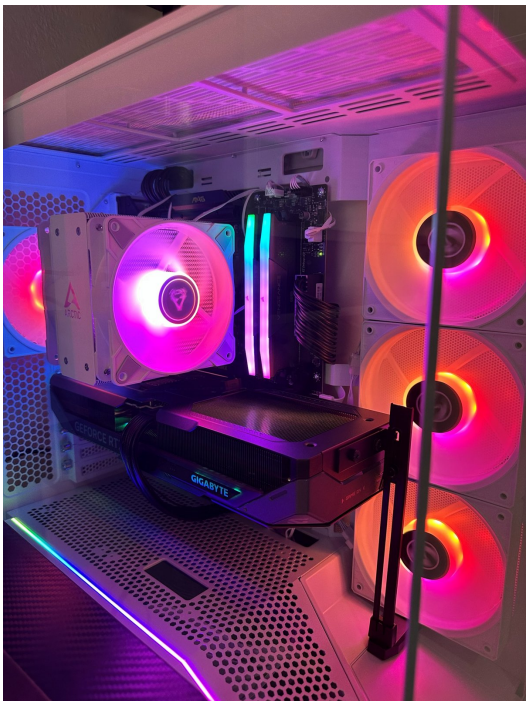


Figure 13. *Pink lighting. Photograph IMG_4178.*



Figure 14. *Green lighting. Photograph IMG_4179.*

11. Lessons Learned and Future Upgrades

11.1 Lessons Learned

- **Verify the Mounting Hardware Before Tightening.** Sorting the cooler's Intel and AMD screws before starting would have prevented the uneven mount described in Section 3.2.
- **Check Header Labels Against the Manual.** A header-by-header check before first power-on caught the cooler cable on the wrong header (Section 3.4).
- **Keep a Second Display Path.** Integrated graphics turned a blank screen into a working route to UEFI and Windows Setup (Section 4.1).
- **Stage Drivers Offline.** Labeled driver folders on a separate USB drive removed any dependence on Windows finding network drivers on its own.
- **Change One Variable at a Time.** Limiting early tuning to EXPO keeps any future instability traceable to a single change.
- **Look for Every Layer of a Setting.** Windows can keep legacy controls alongside newer ones, as the screen saver's sign-in checkbox showed (Section 8.3).
- **Read Sensor Counters in Context.** One counter can look alarming on its own; the related error counters and WHEA determine whether it matters (Section 9.2).
- **Record Versions as I Go.** A screenshot of each version at install time creates a record that later updates cannot overwrite, instead of reconstructing versions from release dates afterward.

11.2 Near-Term Plans

Item	Plan	Status
External Backup	A 4 TB external drive for system images, VM backups, lab projects, and documents.	Deferred until there is data worth protecting
UPS	A 1,000–1,500 VA pure sine wave unit for outages and brownouts.	Not needed yet; the 4,000 J surge protector is in place
BIOS Update	Review MSI's newer releases for AGESA, stability, and security fixes, then update with M-FLASH. The Flash BIOS Button provides a recovery path.	Planned after an extended stability period
CPU Tuning	Precision Boost Overdrive with Curve Optimizer, tested for stability.	Deferred
Extended Validation	Longer CPU, GPU, and memory stress tests with HWiNFO64 logging.	Deferred

11.3 Long-Term Upgrade Path

The B850 platform leaves room to upgrade components individually instead of replacing the PC. My rule is to upgrade a component when it becomes the bottleneck, not when a new generation launches. The timelines below are planning estimates.

Component	Planning Estimate	Upgrade Trigger	Upgrade Path
Storage	Evaluate in 2028–2030	Lab VMs and games approach the 2 TB limit	Add a 4 TB or larger NVMe drive. M2_2 (PCIe 5.0 x4, CPU) and M2_4 (PCIe 4.0 x4, chipset) run at full width; M2_3 is PCIe 4.0 x2 and shares lanes with PCI_E3. [13]
Graphics Card	4–6 years as the primary GPU	Lowering settings to hold frame rates at QHD	Replace the card in PCI_E1 (PCIe 5.0 x16) and confirm its power requirement against the 850 W ATX 3.1 PSU.
Memory	5–8 years or more	Sustained 90–100% use while running multiple VMs	Move to 128 GB, preferably as a matched 2 × 64 GB kit. The board supports up to 256 GB across four slots. [13]
Processor	7–10 years	CPU-bound lab or simulation workloads	A later AM5 processor, if MSI's BIOS support allows a drop-in upgrade.
PSU, Case, and Cooler	8–10 years or more	Fan noise, wear, or a future GPU's power demand	Keep them in service and replace individual fans as they wear.

Appendix A. Sources and References

[1] AMD Ryzen 7 9700X: processor specifications and Figure 1. AMD product page

[2] MSI MAG B850 TOMAHAWK MAX WIFI: slot placement, header labels, debug LEDs, and driver packages; Figure 2 is from the MSI US Store. MSI manual · MSI support · MSI store product gallery

[3] Gigabyte GeForce RTX 5070 GAMING OC 12G: model, dimensions, power connector, recommended PSU, and Figure 3. Product and gallery · Gigabyte specifications

[4] Corsair Vengeance RGB CMH64GX5M2N6000Z40: exact kit listing and Figure 4. Newegg product page

[5] ARCTIC Freezer 36 A-RGB White: model, dimensions, fan arrangement, and installation documentation; Figure 5 is from Newegg. ARCTIC product page · Newegg exact-model listing

[6] Predator GM7 2 TB: drive specifications and the exact-part retailer image used for Figure 6. Predator Storage specifications · Jumbo Computer BL.9BWWR.119 listing

[7] Corsair RM850x ATX 3.1: dimensions, power connections, and Figure 7. Corsair product page

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Glossary

Terms are defined as they are used in this report. Authoritative references are listed in Appendix A.

12V-2x6	A 16-pin power connector (12 power pins and 4 sense pins) that carries power from the PSU to the graphics card. It is the revised successor to 12VHPWR.
24-Pin ATX and EPS	The 24-pin ATX cable powers the motherboard. Each 8-pin EPS cable feeds the motherboard's CPU power circuit.
Adaptive-Sync and VRR	Variable refresh rate (VRR) lets a display match its refresh timing to the GPU's frame delivery within a supported range. Adaptive-Sync is the VESA standard for VRR over DisplayPort.
Above 4G Decoding and Re-Size BAR	Above 4G Decoding lets the firmware map PCIe devices into memory addresses above the 4 GB boundary. Re-Size BAR builds on it so the CPU can access the GPU's full video memory at once instead of in 256 MB windows.
AM5	AMD's current desktop processor platform. Its LGA 1718 socket has the contact pins on the motherboard rather than on the CPU.
ARGB	Addressable RGB lighting, in which each LED can be controlled individually. It uses a 5 V, three-pin header and is not compatible with 12 V, four-pin RGB headers.
ATX and ATX 3.1	ATX defines the form factors and interfaces for PC components. ATX 3.1 is the power-supply specification revision that this PSU follows.
BitLocker	Windows full-volume encryption. It uses the TPM to protect the encryption key and asks for a recovery key if the boot environment changes unexpectedly.
Boot Order and Installation Media	Boot order sets the sequence in which firmware tries boot devices. The Windows installation USB is bootable media containing Windows Setup.
Chipset	Motherboard hardware that connects the CPU platform to additional devices such as USB controllers, storage, and expansion slots.
CL40	A memory timing rating. A CAS latency of 40 means 40 memory clock cycles pass between a column read command and the first data being available.
Core Isolation and Memory Integrity	Core isolation uses virtualization-based security to protect critical parts of Windows. Memory Integrity, its main feature, blocks untrusted or tampered code from running in the Windows kernel.
CPU and CPU Cooler	The central processing unit executes program instructions. A cooler moves heat from the CPU through a heatsink and into the surrounding air.
CSM	Compatibility Support Module. It emulates legacy BIOS booting on UEFI firmware. Disabling it leaves only UEFI boot, which Secure Boot requires.
DDR5 and MT/s	DDR5 is the fifth generation of double-data-rate memory. MT/s counts millions of transfers per second. DDR5-6000 performs 6,000 MT/s using a 3,000 MHz clock, because data moves on both clock edges.
Device Manager	A Windows tool for checking device detection, driver status, and device errors.
DIMM and A2/B2	A DIMM is a memory module. A2 and B2 are the motherboard slots used for this two-module kit, one in each memory channel.

DisplayPort and HDMI	Digital interfaces that carry video and audio to a display. The motherboard and the graphics card each provide their own display outputs.
Driver	Software that lets Windows communicate with hardware such as the chipset, graphics card, and network adapters.
EXPO	AMD Extended Profiles for Overclocking. A profile stored on the memory modules sets their rated speed, timings, and voltage on compatible systems.
Fan Curve, PWM, and Step Timing	A fan curve maps temperature to fan speed. Pulse-width modulation (PWM) controls speed by varying a signal's duty cycle. Step-up and step-down times control how quickly the fan responds to temperature changes.
Flash BIOS Button and M-FLASH	MSI's two firmware update methods. M-FLASH updates the BIOS from within the firmware interface. The Flash BIOS Button updates it from a USB drive with only power connected, which gives a recovery path if an update fails.
Front-Panel Header and USB-C	The front-panel header connects the case's power button, reset switch, and indicator LEDs. USB-C is a connector shape; the motherboard header and cable determine the speed the front USB-C port supports.
fTPM and TPM 2.0	A Trusted Platform Module stores keys and boot measurements in hardware-protected storage. fTPM is AMD's firmware implementation inside the processor. Windows 11 requires TPM 2.0.
G-SYNC Compatible	NVIDIA's certification for variable-refresh displays that pass its compatibility testing with GeForce GPUs.
GPU and GDDR7	The graphics processing unit renders and processes graphics. GDDR7 is the dedicated graphics memory on the RTX 5070.
HWiNFO64	A Windows hardware monitoring utility that reports temperatures, clocks, power, fan speeds, and error counters from system sensors.
Hyper-V and Windows Sandbox	Hyper-V is the hypervisor built into Windows for running virtual machines. Windows Sandbox is a disposable, isolated Windows desktop for testing untrusted files. Both are included in Windows 11 Pro.
Integrated Graphics and UMA	The Ryzen CPU includes a built-in GPU that drives the motherboard's display outputs. Unified Memory Architecture (UMA) lets it use part of system RAM as graphics memory.
Local Group Policy	A Windows Pro administrative tool (`gpedit.msc`) for configuring system, security, and user policies on a single PC.
M.2 2280, NVMe, and SSD	An SSD stores data in solid-state flash memory. M.2 2280 is its 22 × 80 mm form factor. NVMe is the storage protocol it uses over PCIe.
PBO and Curve Optimizer	Precision Boost Overdrive lets a Ryzen processor boost beyond stock power limits when temperature and power allow. Curve Optimizer adjusts each core's voltage-frequency curve. Both require stability testing.
PCIe Gen4 x4 and Link Recovery	PCI Express Gen4 x4 means four fourth-generation lanes. A link recovery count records how often a link entered its recovery state, during which it can retrain.
PEG	PCI Express Graphics. Selecting PEG as the initial display adapter tells the firmware to prefer the graphics card in the PCIe slot.

POST and EZ Debug LED	The power-on self-test is the firmware's startup hardware check. MSI's EZ Debug LEDs (CPU, DRAM, VGA, and BOOT) indicate which stage the system is in or where it stopped.
PSU and Modular Cabling	The power supply unit converts wall power into the DC voltages the PC uses. A fully modular PSU has detachable cables for every connection.
QHD and Refresh Rate	QHD is a resolution of 2560 × 1440 pixels. Refresh rate, in hertz (Hz), is how many times per second the display can update its image.
Reputation-Based Protection	Windows Security features, built on Microsoft Defender SmartScreen, that warn about or block malicious or potentially unwanted apps, files, and websites based on their reputation.
RGB Full and Bits per Channel	Full-range RGB uses all digital code values (0–255 at 8 bits) for red, green, and blue. Eight bits per channel give each color channel 256 levels.
Secure Boot	A UEFI feature that allows only boot software signed by keys the firmware trusts to run.
Standoff and GPU Support	A standoff holds the motherboard off the case tray. A GPU support post carries part of the graphics card's weight to limit sag.
SVM	Secure Virtual Machine, AMD's hardware virtualization extension (AMD-V). Enabling it in firmware lets hypervisors such as VirtualBox use hardware-assisted virtualization.
Thermal Compound and Throttling	Thermal compound fills microscopic gaps between the CPU and cooler to improve heat transfer. Thermal throttling reduces a component's speed to keep it within temperature limits.
UEFI and BIOS	UEFI is the modern firmware interface that initializes hardware and starts the operating system. BIOS is the older firmware standard; the name is still commonly used for the firmware setup screen.
UPS	Uninterruptible power supply. A battery backup that keeps the PC running through short outages and allows a clean shutdown during longer ones.
WHEA	Windows Hardware Error Architecture. It records hardware error reports, including corrected errors that never cause a crash.
Windows Activation	Microsoft's process for validating an installed Windows edition against a product key or digital license.
Windows Hello PIN	A sign-in PIN tied to this specific device and protected by the TPM. Unlike a password, it cannot be used to sign in from another PC.